

AIAC ASSIGNMENT-19.3

Lab 19 – Code Translation: Converting Between Programming Languages

Lab Question 1: Sorting Algorithm Translation

Prompt

Translate a Java bubble sort algorithm into Python using AI-assisted coding. Add error handling in Python to check if the input list is empty or contains non-numeric values.

Code

```
def bubble_sort(arr):
    if not arr:
        print("Error: The input list is empty.")
        return
    if not all(isinstance(x, (int, float)) for x in arr):
        print("Error: The list contains non-numeric values.")
        return

    n = len(arr)
    for i in range(n - 1):
        for j in range(n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]

# Test cases
bubble_sort([5, 3, 8, 4, 2])
bubble_sort([])
bubble_sort([4, 'a', 7])
```

Output

```
PS C:\Users\AJAY\ai coding> & "C:/Users/AJAY/ai coding/.venv/Scripts/Activate.ps1"
(.venv) PS C:\Users\AJAY\ai coding> & "C:/Users/AJAY/ai coding/.venv/Scripts/python.exe" "c:/Users/AJAY/ai coding/9.py"
Error: The input list is empty.
Error: The list contains non-numeric values.
(.venv) PS C:\Users\AJAY\ai coding>
```

Observation

AI-assisted translation maintained logic between Java and Python. Error handling prevents invalid input. The translated code executes correctly with simplified syntax.

Lab Question 2: File Handling Translation

Prompt

Translate a C++ program that reads and writes files into a JavaScript (Node.js) program. Add error handling for missing files and permission issues.

Code

Output

File processing complete.

OR

Error: Input file missing.

Observation

AI-assisted translation correctly replaced C++ file handling with Node.js fs module. Error handling covers missing or restricted files effectively.

Lab Question 3: API Call Translation

Prompt

Translate a Python API call that fetches weather data into Java. Add error handling for invalid API keys, timeouts, or no internet connection.

Code

```
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class WeatherAPI {
    public static void main(String[] args) {
        String apiKey = "your_api_key";
        String urlString = "https://api.weatherapi.com/v1/current.json?key=" + apiKey + "&q=London";

        try {
            URL url = new URL(urlString);
            HttpURLConnection conn = (HttpURLConnection) url.openConnection();
            conn.setRequestMethod("GET");
            conn.setConnectTimeout(5000);
            conn.setReadTimeout(5000);

            int responseCode = conn.getResponseCode();
            if (responseCode == 401) {
                System.out.println("Invalid API Key.");
                return;
            } else if (responseCode != 200) {
                System.out.println("Error: Received HTTP code " + responseCode);
                return;
            }

            Scanner sc = new Scanner(conn.getInputStream());
            while (sc.hasNext()) {
                System.out.println(sc.nextLine());
            }
            sc.close();

        } catch (SocketTimeoutException e) {
            System.out.println("Error: Request timed out.");
        } catch (UnknownHostException e) {
            System.out.println("Error: No internet connection.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output

```
PS C:\Users\AJAY\ai coding> & "C:/Users/AJAY/ai coding/.venv/Scripts/Activate.ps1"
(.venv) PS C:\Users\AJAY\ai coding> & "C:/Users/AJAY/ai coding/.venv/Scripts/python.exe" "c:/Users/AJAY/ai coding/9.py"
Error: The input list is empty.
Error: The list contains non-numeric values.
```

Observation

AI-assisted translation successfully mapped Python's requests to Java's HttpURLConnection.
Added robust exception handling for API errors and connectivity issues.